NAVI

Navigation Assistant for Visitor Interaction

An autonomous follow-me micro aerial vehicle for outdoor campus wayfinding

Comprehensive Project Report

Design decisions, campus data analysis, software delivered, and outstanding measurements. Prepared for Review 1 of the interdisciplinary project.

Institution	Vellore Institute of Technology, Chennai
Schools	SCOPE (Computer Science), SENSE (Electronics), SMEC (Mechanical)
Laboratory	ADAPT, AB3-710
Team size	Five members
Clearance	Campus flight permission granted for all campus coverage; project operates under ADAPT's existing registration.

Status of the figures in this report. Every campus measurement is computed from the surveyed dataset and is reproducible. Every aircraft performance figure is an engineering estimate and is labelled as such. Section 16 lists precisely which is which. No flight testing has taken place.

1. Project Identity

The project is titled **NAVI — Navigation Assistant for Visitor Interaction**, with the subtitle *An Autonomous Follow-Me Micro Aerial Vehicle for Outdoor Campus Wayfinding*.

The acronym was selected over alternatives because it functions as a legitimate expansion, is short enough to say aloud, and carries an existing association with a small guiding companion. The term *micro aerial vehicle* is used in place of “mini drone” throughout: it is the standard term in the literature and aligns the project with the sub-250 g regulatory vocabulary even though the chosen airframe is heavier.

2. Problem, Relevance and Related Work

2.1 The problem

First-time visitors to a large campus routinely get lost. Three existing approaches each fail for a structural reason rather than a quality reason:

- **Signage is passive.** Boards show a map, not a route. The visitor must establish their own position on that map before it becomes useful.
- **Human guides do not scale.** Staff and student volunteers are finite, and demand spikes on event days precisely when they are least available.
- **Consumer navigation applications stop at the gate.** They carry little or no internal campus pathway detail.

The common failure is **self-localisation**. Every solution above requires the visitor to work out where they currently are and translate a route representation into what they can see. An aerial guide the visitor simply

follows removes that step entirely, and is additionally unobstructed by crowds, parked vehicles and ground clutter.

2.2 Related work: the existing campus wayfinder

VIT Chennai already operates an official 2D/3D interactive navigator at chennaiwayfinder.vit.ac.in, credited to Madras MindWorks. It carries a complete, maintained database of campus buildings and facilities with detailed 3D geometry and category search, and is accessible from any browser.

This must be raised by the team before a reviewer raises it. The honest position is that NAVI is complementary rather than competing: the wayfinder still presents a map, and therefore still requires the self-localisation step identified above. The wayfinder's dataset subsequently became the primary data source for this project, described in section 8.

3. Scope Decisions and How They Were Reached

The scope was not fixed at the outset. It narrowed through a sequence of decisions, each forced by a constraint rather than chosen for convenience. Recording the sequence is useful because a reviewer asking “why this scope” gets a derivation rather than an assertion.

Leading, not pointing	The drone physically leads the visitor rather than projecting or announcing directions from a fixed hover. This is a substantially harder engineering problem and is the source of the project's technical contribution.
Outdoor, not indoor	Indoor flight would require UWB anchors, fiducial markers or visual-inertial odometry, since GNSS is unavailable. The available laboratory hardware is RTK-equipped and the airframes are too large to fly safely at head height indoors.
Roads as well as footpaths	Broadens coverage and improves RTK sky view, but introduces overhead cabling, tree canopy and vehicle traffic as new constraints.
Fully autonomous flight	Flight is autonomous; a safety pilot is retained during development and tuning as sensible practice rather than as a compliance requirement, since clearance has been granted.
Pre-inspected routes	Routes are generated from a surveyed graph and inspected before flight. The drone never improvises a path. A whitelist check rejects any waypoint that does not correspond to a surveyed node.
Segment-level altitude	Cruise height is set per route segment from the surveyed worst overhead obstruction on that segment, not as a single campus-wide value.

3.1 Scope statement

In scope. Outdoor flight over surveyed campus roads and pathways; guided legs of at most 350 m sized for a 0.8 m/s walking speed; a fixed set of pre-mapped destinations with every route pre-generated and inspected; one visitor or one small group at a time; daytime operation in fair weather below a defined wind threshold; longer routes staged across surveyed launch pads with battery swaps; fully autonomous flight under institutional clearance; no flight directly over occupied vehicles or live traffic lanes.

Out of scope. Indoor navigation and building interiors; beyond-visual-line-of-sight operation; multiple simultaneous guided visitors; night flight and adverse weather; autonomous charging docks with precision landing; carrying luggage or any payload for the visitor; live obstacle detection and dynamic in-flight re-routing.

4. Multidisciplinary Basis and Team

No single discipline can deliver a drone that safely leads a human being across a live campus. Safety and human factors are not additions to the engineering; they constrain it.

Mechanical	Airframe integration, propulsion, vibration isolation, weight and thrust budget.
Electronics and Communication	RTK GNSS correction link, telemetry, sensor integration, power distribution.
Computer Science	Route graph, path planning, MAVLink mission logic, pacing controller, web interface.
Human Factors	How a stranger reads a flying guide: pacing, lead distance, arrival cues, trust.
Safety and Regulation	Geofencing, failsafe design, risk assessment for shared campus roads.
Survey and Mapping	Pathway and road surveying, junction geo-referencing, overhead obstruction logging.

4.1 Role allocation

Member	Registration	School	Role	Contribution
Carollo Aradhan V	26BMH1173	SMEC	Flight platform lead	Airframe assembly and integration, propulsion and payload mounting, vibration isolation, weight and thrust budget.
Utkarsh Dilipkumar Ghorpade	26BEC1215	SENSE	Positioning and electronics lead	RTK base and rover setup, correction link, telemetry, sensor integration, power distribution.
N Arun	26BCE1550	SCOPE	Autonomy and flight software lead	MAVLink mission scripting, companion computer integration, pacing controller, failsafe logic.
Muhammed Farhan	26BCE1551	SCOPE	Mapping and path planning lead	Campus route graph, RTK survey data processing, shortest-path planner, per-segment altitude model.
Sayanjeet Das	26BCE5502	SCOPE	Interface and validation lead	Visitor web interface and kiosk, destination database, guided trials, documentation and reporting.

Three of the five members are from Computer Science. If questioned, the defence is that the software workload divides cleanly into three independent areas — flight autonomy, mapping and planning, and the visitor-facing system — and each contribution above is distinguishable from the others.

5. Hardware Platform and Laboratory Inventory

The project requires no significant procurement. All primary hardware is already held in the ADAPT laboratory. This materially strengthens the feasibility case: the project is not blocked on purchasing, and the RTK equipment in particular is what makes the outdoor approach viable at all.

Airframe	S500 quadcopter with Pixhawk 2.4.8 flight controller, or an X500 V2 build kit.
Positioning	Helical H-RTK F9P base and rover pair; Here4 multiband RTK GNSS.
Compute	Pixhawk 6X with Jetson Orin NX bundle; Raspberry Pi 5 as an alternative.
Perception	Intel RealSense D455 depth camera; SIYI A8 gimbal with object tracking.
Communication	SiK telemetry at 915 MHz; Skydroid T12 / H12 control link.
Power	Multiple LiPo packs, SkyRC charger, discharger and BD250 battery analyser.
Visitor interface	Raspberry Pi 7-inch touchscreen for the gate kiosk.

Anticipated procurement is limited to propeller guards, mounting hardware and consumables.

5.1 Alternatives considered and rejected

A ready-to-fly programmable platform such as a Tello was evaluated as the lowest-cost route, at roughly INR 10,000 against INR 20,000-30,000 for a custom build. It was rejected once the laboratory inventory was known: the available hardware is both free to the project and substantially more capable. A larger-airframe efficiency rebuild on the X500 kits was also considered and deferred; it would extend endurance to an estimated 18–25 minutes but costs weeks of assembly and tuning, and the launch-pad architecture in section 7 solves the same problem at no cost.

5.2 Engineering constraints on the platform

- Hover throttle must remain below approximately 60 per cent; above this the aircraft is overloaded, with poor endurance and poor wind handling.
- Vibration on all three axes must stay within ArduPilot's acceptable band. A companion computer and camera change the vibration profile, and excessive vibration causes the EKF to reject GPS mid-flight. Soft-mounting is required.
- The companion computer must have its own BEC rather than drawing from the flight controller's power rail, to avoid browning out the Pixhawk in flight.
- The 5200 mAh 4S packs in the inventory are not recommended: 920 kV motors on 4S drive 1045 propellers past their comfortable range, and motors and ESCs will run hot.
- The 6S packs of 8000–22000 mAh belong to the Tarot octacopter and are not applicable; a 22000 mAh 6S pack alone weighs more than the entire S500.

6. Endurance Analysis

This section derives the single most important design number in the project. It is reproduced in full because the derivation, not the number, is what a reviewer should be shown.

6.1 Baseline endurance

The S500 with a 3300 mAh 3S pack, 2212/920 kV motors and 1045 propellers is estimated to draw 12–16 A in hover at an all-up weight of 1.2–1.4 kg. Using 80 per cent of pack capacity, this gives approximately 10–11 minutes bare, falling to about 8 minutes once roughly 300 g of companion computer, camera and mounting hardware is added.

6.2 Walking speed

The intended users are visiting parents and guests rather than students. Typical adult walking speed falls with age, clustering around 1.0–1.2 m/s for older adults. Designing to that median would leave approximately half of the intended users short of battery partway through a guided walk, and the failure mode is not graceful: it is a low-battery return-to-launch during someone's guided walk.

The endurance budget is therefore sized at **0.8 m/s**, with 1.0 m/s quoted as the expected typical case. A 15 per cent dwell overhead is added for stopping, looking around and asking questions, and 25 per cent battery reserve is held back.

6.3 Distance iterations

Several candidate maximum distances were tested against this budget. The table below shows the endurance required for a one-way guided route with the drone landing at the destination rather than returning.

Route length	At 0.8 m/s	At 1.0 m/s	At 1.3 m/s	Verdict against ~11 min available
1500 m	~45 min	~36 min	~28 min	Not feasible; also outside visual line of sight
700 m	21.0 min	16.8 min	12.9 min	Not feasible at any walking speed
600 m	18.0 min	14.4 min	11.1 min	Feasible only for a brisk walker
460 m	13.8 min	11.0 min	8.5 min	Feasible at 1.0 m/s
350 m	10.5 min	8.4 min	6.5 min	Feasible at the 0.8 m/s design floor

6.4 Result

Maximum guided leg: 350 m. This follows from an estimated 8 minutes of flight with payload, a 0.8 m/s design walking speed, a 15 per cent dwell allowance and a 25 per cent battery reserve. It is a derived figure, not a chosen one. Should the pack capacity test in section 17 return lower than rated capacity, this number falls proportionally and the launch-pad result in section 7 must be recomputed.

7. Launch Pad Architecture

A 350 m leg limit does not by itself bound total guided distance. Chaining legs between surveyed launch pads does, and it converts the constraint from “how far can the aircraft fly” into “how are the pads placed” — a question the team controls.

7.1 Tiers considered

Tier 1 — surveyed pads	A marked, RTK-surveyed spot at several points around campus. No hardware: a painted circle and a coordinate in the graph. A team member carries the aircraft and spare packs. Cost: negligible. **Adopted.**
Tier 2 — staffed swap points	The same pads plus a weatherproof box of charged packs at each. Estimated INR 2,000-5,000. Available as an upgrade.
Tier 3 — autonomous docks	Precision landing, contact charging, weatherproof enclosures and mains power at each site. Estimated INR 80,000 and a semester of work. **Out of scope**, listed as future work.

7.2 Optimisation result

Pad placement was solved as a set-cover problem over the real campus graph: find the minimum number of nodes such that every destination lies within one 350 m leg of some pad. A greedy solution returns six pads.

Pad	Node	Latitude	Longitude
Gate Pad	111	12.840649	80.153159
Central Pad	50	12.843086	80.154410
East Pad	40	12.841788	80.155996
North-West Pad	7	12.843654	80.152814
South Pad	27	12.840475	80.154658
E Block Pad	68	12.844735	80.160818

With these six pads, no destination on campus requires more than one leg. The longest single leg anywhere is 316 m, to VIT-SEDAXIS, which sits inside the 350 m limit with margin. This is the strongest single result the project currently holds, and it was computed from the real campus network rather than estimated.

8. Campus Data Acquisition

8.1 The wayfinder path graph

The campus wayfinder exposes its routing graph at `/data/path.json`. It contains 278 waypoints with latitude and longitude, per-node connection lists, 36 building entrances mapped to waypoint identifiers, and a set of one-way road constraints. This replaced every placeholder coordinate previously used in the project software.

Two nodes (276 and 277) carry blank latitude and longitude in the source. The file also stores a local metre grid whose origin is node 0; the axes were confirmed to be east and south at one unit per metre by cross-checking a node with both representations, and both blank nodes were reconstructed exactly from their grid coordinates.

The one-way road constraints apply to vehicles and are ignored for flight, but are retained in the dataset in case visitor routing by vehicle is ever added.

8.2 The campus boundary

The campus boundary was taken from OpenStreetMap way 336095563, tagged `amenity=university` and named Vellore Institute of Technology — Chennai. It is a closed way of 36 vertices, last edited in January 2026 and therefore actively maintained.

8.3 Licensing

OpenStreetMap data is licensed under ODbL 1.0. The attribution “Campus boundary © OpenStreetMap contributors, ODbL 1.0” appears in the application status bar, on the boundary layer tooltip, and in every exported mission file. The share-alike terms apply to any derived database, which should be noted in the project documentation rather than discovered at a later review.

9. Graph Analysis Results

Nodes	278 surveyed waypoints in the source; 260 retained for flight after filtering.
Segments	303 undirected segments; 8.24 km of total path network.
Segment length	Minimum 1.4 m, median 20.8 m, maximum 383.7 m.
Connectivity	A single connected component. Node 0 is a datum marker with no connections and is excluded.
Campus extent	42.2 hectares; approximately 991 m north-south by 1236 m east-west; 4266 m perimeter.
Destinations	36, all reachable from every pad after filtering.

9.1 Distances from the main gate

Before pad optimisation, routing from the main gate alone gave a median distance of 453 m and a maximum of 1264 m to E Block Hostel. Only 12 of 36 destinations were reachable in a single 350 m leg. This is what demonstrated that a single launch point was never viable, and it is a measured result rather than an assumption.

9.2 The long-segment problem

One segment, between nodes 127 and 182, is 383.7 m in a single hop with no intermediate node. No pad placement can split it. The issue resolved itself when node 182 was identified as lying off campus on the public approach road and removed from the flight graph, but the general point stands: a single segment longer than the leg limit cannot be fixed by pad placement and must be fixed by adding a node during survey.

9.3 Off-campus filtering

Seventeen nodes were removed as genuinely off campus: nodes 1–6, 108 and 179–182 lie 147–461 m outside the boundary along Vandalur–Kelambakkam Road to the west, and nodes 242–247 lie 75–336 m outside to the south. These are public approach roads the wayfinder includes so it can route people in from outside. They are not campus paths and are not the project's to fly over. After removal the graph remains a single connected component with all 36 destinations reachable.

10. Geofence Analysis

Cross-checking the wayfinder graph against the OpenStreetMap boundary served two purposes: defining the geofence, and verifying the coordinates against an independent source.

10.1 Findings

Of the 278 nodes, 37 fall outside the boundary polygon. Seventeen are genuinely off campus, as described in section 9.3. The remaining 26 sit 2–40 m outside and are real campus paths near the perimeter; the discrepancy reflects the precision of a hand-traced boundary rather than an error in the graph.

10.2 Consequence for the geofence

The OpenStreetMap polygon must **not** be used directly as the flight geofence: it would reject legitimate perimeter paths. The implemented rule is that a breach occurs when a position lies outside the polygon by more than **40 m**. This is implemented as ray-casting point-in-polygon combined with point-to-segment distance, which avoids offsetting the polygon geometrically. The buffer figure is derived from the observed discrepancy and should be stated with that justification.

10.3 Coordinate verification

The cross-check is itself a result. The wayfinder coordinates agree with OpenStreetMap to within a few tens of metres with no systematic offset visible; every outlier is explained by geometry rather than by a datum error. This does not replace RTK verification, which remains outstanding, but it establishes that no gross error is present.

10.4 Boundary risk points

Twelve nodes lie within 25 m of the boundary and nine within 4 m, with node 169 essentially on the line. These are the positions at which a wind gust is most likely to trigger a geofence event. They are rendered on their own map layer in the application and should be flagged in the graph so the pacing controller holds further from the edge on those segments. All six launch pads sit inside the boundary at 33–190 m clearance; the Gate Pad at 33 m is the tightest, which is expected for a gate.

11. System Architecture

```
Visitor phone  ->  Ground station  ->  Companion computer  ->  Pixhawk
  destination        route planning       mission logic and       position targets,
  choice, live       over the campus      pacing controller;      stabilisation and
  position via       graph; RTK base      MAVLink to the          all failsafes
  Geolocation API    streams RTCM3        flight controller
```

11.1 Key design decision: phone GPS, not onboard vision

The visitor is tracked by their own phone's GPS rather than by onboard computer vision. Re-identifying one specific pedestrian from altitude, in variable sunlight, among other people is an open research problem. Phone GPS is accurate to a few metres, which is sufficient for pacing. This converts a research problem into an engineering one and is the decision most likely to determine whether the project completes. Onboard depth and gimbal cameras are retained as a secondary confirmation channel.

11.2 Key design decision: per-segment altitude

Cruise altitude is derived from the surveyed worst overhead obstruction on each route segment rather than set as a single campus-wide value. On road corridors this is what makes the design safe: campus roads typically carry power, streetlight and data cabling strung between poles, and mature trees arching over the carriageway. Both are collision surfaces, and canopy additionally causes GNSS multipath.

11.3 Pacing control

- The guest's phone posts position at approximately 1 Hz to the ground station over HTTP.
- The controller computes **along-route** distance between guest and drone, not straight-line distance. Both positions are projected onto the route polyline; otherwise a guest cutting a corner registers as a large gap.
- Target lead distance is approximately 10–12 m, to be tuned from testing. Too close is intimidating; too far and the visitor loses sight of the aircraft.
- If the gap is below threshold the next position target is released; if above, the drone holds. Hysteresis is applied, otherwise the drone oscillates between advancing and holding.
- State machine: IDLE, ARMING, TAKEOFF, EN_ROUTE, PACING, HOLDING, VISITOR_LOST, RETURNING, LANDING, COMPLETE, FAILSAFE.

11.4 Zero-install visitor interface

No application is installed on the visitor's phone. A QR code opens a web page served from the ground station, and the browser Geolocation API supplies position after a permission prompt. Three practical constraints apply: HTTPS is mandatory, since browsers refuse geolocation over plain HTTP; both Android and iOS throttle or suspend `watchPosition` when the screen locks, which is the most likely cause of a failed demonstration; and the phone must be able to reach the ground station on the network. A dedicated device carried by the visitor remains available as a fallback and is defensible in a report as a controlled-trial choice.

12. Software Delivered

Three working applications were produced. All run as single HTML files with no build step, no server and no installation, using Leaflet 1.9.4 from CDN.

Visitor kiosk	Destination search and selection, Dijkstra routing, automatic leg splitting at the 350 m limit, walking distance and time at 0.8 m/s, and generation of the waypoint mission. Intended for the gate touchscreen.
Mission control dashboard	Waypoint planning with place, drag, reorder and per-waypoint altitude and speed; route distance, duration, altitude range and battery estimates; zone conflict detection sampled every 15 m along each leg; flight simulation with live telemetry and flown trail.
Full NAVI application	A single-page application with a landing page, ten operator pages and a separate visitor mode. This is the primary deliverable.

12.1 The full application

- **Pages.** Overview, Live Map, Route Planner, Mission Control, Fleet, Simulation, Safety, Destinations, Statistics, About and Settings, with a collapsible sidebar.
- **Themes.** Dark as the primary operations mode and light as a daytime planning mode, each designed rather than inverted, persisted in browser storage.
- **Simulation.** The visitor walks the route at 0.8 m/s and the drone paces to hold a 10 m lead, rate-limited so the gap closes and opens realistically. Six fault injections — visitor stopped, visitor delayed, RTK loss, telemetry loss, low battery and geofence breach — each drive the safety state machine to the documented response.
- **Data.** 260 nodes, 36 destinations, 6 pads and the 36-vertex boundary, embedded as a JSON literal and verified for dangling connections and unreachable destinations.
- **Export.** Mission JSON with per-node coordinates, altitudes, provenance notes and attribution, labelled as a simulation export.

12.2 Algorithms

Segment length	Haversine, computed at runtime. Nothing is precomputed, so editing a coordinate updates distance, time and battery estimates everywhere.
Routing	Dijkstra over the adjacency list, from the selected pad to the nearest entrance of the chosen destination.
Leg splitting	Greedy: accumulate segments until the 350 m limit would be exceeded, then break at the last node before it.
Geofence	Ray-casting point-in-polygon with point-to-segment distance; breach when outside the polygon by more than 40 m.

12.3 Telemetry abstraction

```
Telemetry.use(provider)   // provider emits a state object
Telemetry.on(callback)    // every UI element subscribes

Default: SimulatedTelemetry
Future:  MAVLink | WebSocket | MQTT  — no UI code changes required
```

12.4 Defects found and fixed

Two genuine defects were found by testing. First, a startup crash: the element `<div id="map">` causes browsers to create a global named `window.map` pointing at the DOM element, so a guard intended to test for the Leaflet instance was always true and read a `let` binding before its declaration. The resulting ReferenceError killed the whole script. Second, an unknown destination name crashed route planning. Both are fixed; a global error handler now surfaces failures rather than failing silently.

12.5 Hosting

The static application should be hosted on GitLab Pages: the project already uses GitLab, it is free, and it provides HTTPS automatically — which matters because the visitor's phone will not share GPS over plain HTTP. Static hosting cannot receive the visitor's position, so the architecture splits: the dashboard and kiosk map are served publicly and statically, while position intake and mission upload run on a small local server on the ground station. This split is arguably correct regardless, since the flight-critical path should not depend on internet reachability mid-flight.

13. Safety Envelope

Condition	Action
RTK fix lost beyond timeout	Hold position, then return to launch
Telemetry link lost	Return to launch
Visitor position stale beyond 30 s	Hold position, then return to launch
Battery below reserve	Return to launch immediately
Geofence breach	Stop and hold

13.1 Design principles

- The safe default is always to hold and then return, never to continue on stale data.
- Waypoints are whitelisted: any target not matching a surveyed node is rejected before upload. This is a cheap implementation and a strong answer to “how do you guarantee it will not fly somewhere unexpected”.
- Routing prefers verges and open ground over carriageway wherever an alternative exists, implemented as a cost multiplier on road-surface edges in the planner.
- The visitor walks the footpath or road shoulder, not the carriageway; routing a pedestrian into traffic behind a drone would create a hazard worse than the one solved.
- Safety state machine: SAFE, WARNING, HOLD, RETURN_TO_LAUNCH, EMERGENCY.

13.2 On the safety pilot

Autonomous flight and a safety pilot are not in conflict. The flight is autonomous throughout; the safety pilot holds a transmitter and does nothing unless something goes wrong. Published work on fully autonomous UAV systems is routinely flown this way. Since campus clearance has been granted, retaining a safety pilot during Phase 3 tuning is sensible practice rather than a compliance requirement, and it costs the project nothing.

14. Documents Produced

Document	Format	Purpose
Review 1 presentation (v1, v2, v3)	PPTX	Progressive versions; v3 has 20 slides against the Review 1 rubric
Review 1 presentation (dark redesign)	PPTX	v3 content in a dark, varied-layout visual system
Review 1 content listing	PDF, TXT	The same content as plain numbered sections
Technical glossary	PDF	~40 terms defined for a beginner audience, in nine sections
Technical specification and review Q&A;	PDF	Subsystem breakdown plus six prepared answers
Project and software summary	PDF	Project overview and website technical specification
Visitor kiosk	HTML	Gate touchscreen application
Mission control dashboard	HTML	Ground station planning and simulation
Full NAVI application	HTML	Primary software deliverable
This report	PDF	Comprehensive record of the work

Content across the presentation, the PDF listing and the text file is identical; any edit must be applied to all three.

15. Work Plan and Verification Gates

Gates matter more than dates at this stage: they demonstrate that the team knows what "done" means for each phase. Durations are expressed as proportions because the project timeline has not been fixed.

Phase	Work	Gate
0 — Clearances	Confirm registration status, department permission, licensed pilot, and whether campus roads carry overhead cabling.	Written permission held; cabling question answered
1 — Survey (~20%)	RTK walkthrough of roads and pathways; log overhead obstructions and fix quality per segment; site the pads.	Graph covers every destination; pads sited so no leg exceeds 350 m
2 — Waypoint flight (~25%)	Bench test, manual hover, then autonomous node-to-node routes in open ground with no visitor.	Stable hover, then repeatable route flight without intervention
3 — Human pacing (~30%)	Phone position sharing; pacing loop; lead distance and hold tuning; edge cases.	Pacing validated in SITL simulation before any live flight
4 — Interface and validation (~25%)	Kiosk and web page; failsafe testing; guided trials with volunteers.	Every failsafe triggered deliberately and observed

15.1 Bring-up sequence

Most student drone projects fail during bring-up rather than in software. Assume nothing in the laboratory works until proven: inspect the airframe, capacity-test every pack, bench-test with propellers removed, calibrate everything from scratch rather than trusting existing settings, and achieve a steady manual hover before any autonomy is layered on. Test the pacing loop in ArduPilot SITL, where the code can crash a hundred times for free, before it touches the real aircraft.

16. Verification Status

This section exists so that no figure in the project is quoted with more confidence than it deserves.

16.1 Computed and reproducible

- Campus area 42.2 ha, perimeter 4266 m, extent 991 m by 1236 m.
- Path network 8.24 km across 303 segments; median segment 20.8 m.
- 260 flight-eligible nodes; 36 destinations; single connected component.
- Distances from the main gate: median 453 m, maximum 1264 m.
- Six-pad set cover; longest leg anywhere 316 m; no destination needs two legs.
- 37 nodes outside the boundary polygon; 12 within 25 m of it.
- All routing, leg splitting and geofence results in the application.

16.2 Estimated, not measured

Approximately 8 minutes of flight with payload. Derived from published motor and propeller characteristics, not from a timed flight. Every downstream figure — the 350 m leg limit, the six-pad result, battery percentages in the application — depends on it.

0.8 m/s design walking speed. Chosen from general adult walking-speed literature, not from measuring the intended users.

Model-derived coordinates. The wayfinder graph comes from a 3D model, not an RTK survey. Cross-checked against OpenStreetMap with no systematic offset visible, but not verified to centimetre accuracy.

Flat 15 m cruise altitude. A placeholder in the software pending the overhead-obstruction survey.

16.3 Not present at all

- No flight testing of any kind has taken place.
- No camera stream is connected; the application states this rather than showing a placeholder.
- No aircraft telemetry; all telemetry is simulated and labelled.
- No mission history; statistics charts use demonstration data and are labelled.

17. Outstanding Data Requirements

Ordered by how much each blocks progress.

#	Measurement	Why it matters
1	Pack capacity test on the SkyRC BD250, every 3300 mAh pack	Old laboratory packs routinely deliver 60–70% of rated capacity. At 65%, the leg limit falls to roughly 230 m and the six-pad result must be recomputed.
2	Timed hover to failsafe with full payload, plus hover throttle percentage	One measurement replaces the estimate that every other figure depends on.
3	Overhead obstruction survey: worst cabling or canopy height per segment	The only item actively blocking the software. Replaces the flat 15 m placeholder with a per-segment altitude model.

#	Measurement	Why it matters
4	RTK spot-check on five or six known nodes	Establishes whether the wayfinder coordinates carry a systematic offset. If so, one correction applies to everything.
5	RTK fix quality logged per segment during survey	Lets the planner route around dead zones rather than discovering them in flight.
6	Whether campus roads carry overhead cabling or it is underground	Determines whether the survey is a day of GPS logging or a careful obstacle audit.
7	The wayfinder's own location database, if the maintainers will share it	Destination categories in the application are currently inferred from names and may be wrong.
8	Review dates and team availability	The work plan has phases and gates but no calendar.

17.1 First action

Before anything else, ask the ADAPT laboratory in-charge whether the S500 currently flies and whether the RTK pair has been set up before. If both answers are yes, most of the bring-up in section 15.1 can be skipped. If nobody has flown it in two years, Phase 1 changes completely — and that is far better known now than in week six.

18. Risk Register

Risk	Description	Mitigation
Endurance	Roughly ten minutes of flight limits route length.	Map short routes; stage battery swaps between guided runs; six-pad architecture.
RTK fix loss	Signal degrades near tall blocks and dense canopy.	Log fix quality per segment during survey; route around dead zones.
Overhead obstructions	Cabling and tree canopy above road corridors.	Record every span during survey; set cruise altitude per segment.
Traffic below	Campus roads carry vehicles and two-wheelers.	Route over verges and open ground; never over live lanes; visitor walks the footpath.
Wind sensitivity	Slow lead flight is the worst case for an S500.	Characterise position-hold error against wind speed; set an operating threshold.
Phone position noise	Handset GPS varies by device and sky view.	Measure the error distribution across several handsets; apply filtering and threshold hysteresis.
Screen lock	Android and iOS throttle geolocation when the screen locks.	Wake lock, on-screen prompt, and a hold-on-stale-data failsafe; dedicated device as fallback.
Vibration	Companion computer and camera change the vibration profile.	Soft-mount everything; check VIBE values in flight logs before autonomy work.
Capacity shortfall	Packs may deliver well below rated capacity.	Test before Phase 2; fresh 3300–4000 mAh 3S packs are inexpensive if needed.

Regulatory clearance is deliberately absent from this register: campus flight permission has been granted for all campus coverage and the project operates under ADAPT's existing registration.

19. Anticipated Review Questions

The general pattern for every answer: name the uncertainty, state the measurement that resolves it, and state the safe default that applies until then. Reviewers are assessing whether the team understands its own system, not whether every problem is already solved.

“Why not computer vision to follow the person?”	Re-identifying one specific pedestrian from altitude, in variable sunlight, among other people is an open research problem. Phone GPS is accurate to a few metres, which is sufficient for pacing. Effort was spent on the guidance problem rather than an unsolved perception problem.
“What is novel? Waypoint flight is solved.”	Correct, and we do not claim otherwise. The contribution is coupling an autonomous aircraft to the unpredictable pace of a walking human, the surveyed per-segment altitude model, and zero-install visitor guidance. The flight stack is off the shelf by choice.
“The campus already has a wayfinder.”	It does, and it is good. It still presents a map, so the visitor must self-localise. NAVI removes that step. The wayfinder's dataset is our data source.
“Phone GPS is noisy.”	Treated as a measurement problem. We will log position from several handsets on a known route, characterise the error distribution, and set the threshold from that with hysteresis. If the noise is too large for a 12 m lead we widen the lead rather than fight the sensor.

“What if GPS jumps under a tree?”	Single samples are filtered: an implausible walking speed is rejected, and consecutive consistent samples are required before acting. The default response to a large apparent gap is to hold, which is the safe failure direction.
“What if it fails mid-guide?”	Every failure mode has a defined action, each tested deliberately. Refer to the failsafe table in section 13.

Appendix A. Destination Distances

Walking distance along the surveyed graph from the main gate, and from the nearest of the six launch pads. All distances in metres, computed with the Haversine formula over the filtered flight graph.

Destination	From main gate	Nearest pad	From nearest pad	Legs
2nd Gate	346	South Pad	193	1
A Block hostel	509	North-West Pad	118	1
Academic Block 1	380	Central Pad	148	1
Academic Block 2	610	East Pad	184	1
Academic Block 3	500	Central Pad	60	1
Academic Block 4	670	Central Pad	230	1
Academic Block 5	453	East Pad	189	1
Admin Block	86	Gate Pad	86	1
Alpha Block	552	East Pad	127	1
B Block Hostel	548	East Pad	122	1
Basketball Court	252	North-West Pad	202	1
C Block Hostel	684	East Pad	259	1
Cricket Ground	601	Central Pad	142	1
D1 Block Hostel	488	North-West Pad	171	1
D2 Block Hostel	488	North-West Pad	171	1
E Block Hostel	1264	E Block Pad	0	1
Football Ground	286	North-West Pad	155	1
Gazebo	277	South Pad	157	1
Ground	245	North-West Pad	233	1
Guest House	274	North-West Pad	143	1
GymKhaana	413	North-West Pad	22	1
Mahatma Gandhi Auditorium	321	South Pad	127	1
Main Gate	0	Gate Pad	0	1
North Square	483	Central Pad	133	1
Omega Block	360	South Pad	166	1
Sigma Block	556	North-West Pad	165	1
Site Office	382	East Pad	156	1
Staff Parking 1	667	North-West Pad	276	1
Staff Parking 2	570	North-West Pad	178	1

Destination	From main gate	Nearest pad	From nearest pad	Legs
Student's Parking 2	291	South Pad	98	1
Student's parking	167	Gate Pad	167	1
Swimming Pool	465	East Pad	177	1
Tennis Court	374	North-West Pad	106	1
Transport Office	374	East Pad	148	1
VIT-SEDAXIS	742	East Pad	316	1
volleyball Court	211	Gate Pad	211	1

Appendix B. Attribution

Campus boundary geometry: OpenStreetMap way 336095563, © OpenStreetMap contributors, licensed under ODbL 1.0. Campus path graph and building entrances derived from the VIT Chennai Way Finder dataset. Base map imagery: Esri World Imagery and OpenStreetMap, attributed in the application.